

ITA1412 -ETHICAL HACKING

Experiment no:1

Experiment Name: Port Scanning Tools

Procedure:

Step 1: Open Nmap from Kali Linux (Go to Applications->select Information Gathering->select

Nmap)

Step 2: Perform different types of scans9`

(Tcp, Udp, Ack, Syn, Fin, Null, Xmas, Rpc, Idle)- scan types

Scanning Techniques

Flag Use Example

-sS TCP syn port scan nmap -sS 192.168.1.1

-sT TCP connect port scan nmap -sT 192.168.1.1

–sU UDP port scan nmap –sU 192.168.1.1

–sA TCP ack port scan nmap –sA 192.168.1.1

Output:

```
kali@kali: ~  
Session Actions Edit View Help  
zsh: corrupt history file /home/kali/.zsh_history  
(kali@kali)-[~]  
$ ping google.com  
PING google.com (142.251.220.110) 56(84) bytes of data.  
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=1 ttl=64 time=39.3 ms  
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=2 ttl=64 time=45.8 ms  
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=3 ttl=64 time=76.2 ms  
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=4 ttl=64 time=97.1 ms  
64 bytes from pnmaaa-be-in-f14.1e100.net (142.251.220.110): icmp_seq=5 ttl=64 time=108 ms  
^C  
--- google.com ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 4023ms  
rtt min/avg/max/mdev = 39.329/73.263/107.811/27.110 ms  
(kali@kali)-[~]  
$ sudo apt update  
[sudo] password for kali:  
Hit:1 http://http.kali.org/kali kali-rolling InRelease  
731 packages can be upgraded. Run 'apt list --upgradable' to see them.  
(kali@kali)-[~]  
$ sudo apt install nmap  
nmap is already the newest version (7.99+dfsg-1kali1).  
nmap set to manually installed.  
Summary:  
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 731  
(kali@kali)-[~]  
$ nmap --version  
Nmap version 7.99 ( https://nmap.org )  
Platform: x86_64-pc-linux-gnu  
Compiled with: liblua-5.4.8 openssl-3.6.2 libssh2-1.11.1 libz-1.3.1 libpcap-1.10.6 nmap-libdnet  
-1.18.0 ipv6  
Compiled without:  
Available nsock engines: epoll poll select  
(kali@kali)-[~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 08:00:27:5a:87:bc brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 85270sec preferred_lft 85270sec  
    inet6 fd17:625c:f037:2:18ac:924a:5755:8a7b/64 scope global dynamic noprefixroute  
        valid_lft 86253sec preferred_lft 14253sec  
    inet6 fe80::6151:3a26:a3f:762e/64 scope link noprefixroute
```

```
kali@kali: ~  
Session Actions Edit View Help  
$ cat /root/.ssh/config  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
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        valid_lft 86253sec preferred_lft 14253sec  
    inet6 fe80::6151:3a26:a3f:762e/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
  
$ nmap 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 00:00 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000040s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.22 seconds  
  
$ nmap -sV 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 00:01 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.000033s latency).  
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 1000 closed tcp ports (reset)  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 0.54 seconds  
  
$ nmap -p- 127.0.0.1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-20 00:01 -0400  
Nmap scan report for localhost (127.0.0.1)  
Host is up (0.0000020s latency).  
All 65535 scanned ports on localhost (127.0.0.1) are in ignored states.  
Not shown: 65535 closed tcp ports (reset)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.67 seconds  
  
$
```